

OM M. PATEL

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AI Engineer building production AI integrations, data pipelines, and backend systems to help teams make better operational decisions efficiently. **Vector AI Scholar** with about 2 years across AI, data and research roles.

SKILLS

Languages	TypeScript, Python, Java, SQL
ML/NLP/LLM	PyTorch, LangChain, NumPy, QLoRA, OpenAI SDK, Google ADK, Pandas, Scikit-learn.
Backend & Data	React, FastAPI, Flask, Next.js, Airflow, PostgreSQL, Supabase, Neo4j, Redis
Tools	Docker, CI/CD, GitHub, Jira, Vercel, GCP, Playwright, Sentry, Cloudflare, Postman

EXPERIENCE

Applied AI Engineer Jun 2026 – Aug 2026

BarLens, Toronto

- Integrated **vision APIs** (with routing layer) with **Next.js** for extracting data to reduce manual work by **64%**.
- Built **LangSmith** eval layer to measure extraction quality and invoke **HITL validation** and POS reconciliation on user dashboard.
- Designed **Supabase PostgreSQL**-backed data storage, with location-based access and private media retention for stock audit.
- Automated **email ingestion pipeline** to normalize POS sales reports (**EML, CSV, PDF**), **avoiding paid access to POS APIs**.
- Shipped via **AI coding agents: Playwright** e2e tests, **CI** checks and **Sentry** monitoring in production to support reliable releases.

Research Assistant

Mar 2026 – Aug 2026

York University | Prof. Uyen T. Nguyen | Manuscript in preparation

- Built a **Gujarati sentiment-analysis pipeline** across translated SST-2 and native GSAC data to evaluate polarity preservation.
- Achieved **weighted F1 of 0.804** with a **MuRIL**-based model and a 2,908-entry sentiment lexicon on a held-out GSAC test split.

Data Engineer

Dec 2024 – Aug 2025

Uarra, Nigeria (Remote)

- Built Python **ETL pipelines** to parse API logs into a **PostgreSQL** analytics database, serving dashboard endpoints in under **1.5 s**.
- Refactored a Flask monolith into Dockerized **microservices** with **Redis** caching, cutting route latency by **30-40%** (1 s to 600 ms).
- Coordinated with backend and ML teams** to monitor API behavior and support iterative post-deployment improvements.

Data Engineering Intern

Mar 2024 – Jul 2024

Uarra, Nigeria (Remote)

- Developed geocoding APIs with **Flask** and **GeoPandas** to reduce dependency on GMaps API saving **about 350 USD monthly**.
- Scraped and pre-processed **20k+** addresses to create training datasets, producing versioned datasets for ML training.
- Built **MongoDB**-backed credit and **Paystack** payment flows with rate limiting and CSRF protection. Shipped with **Docker** on **AWS**.

PROJECTS

Synapse AI : Teacher Orchestrated Student Learning Platform

May 2026

- Won the AgentShyft Hackathon with a **multi-agent** tutoring platform: agents call course-material tools through an **MCP server** and stream answers over **SSE**. Built the teacher side in **FastAPI** and **Supabase**.

Remote Codex Control : Remote HITL for Coding Agents

Feb 2026 – May 2026

- Built a human-approval layer for autonomous **coding agents**: an operator approves, steers or denies each tool action from a private Telegram chat while execution stays on the local machine.
- Directed AI coding agents to produce 17k+ lines of **TypeScript** with **SQLite** state, a chat allowlist, state-approval protection and a documented threat model.

LLMs for Job-Shop Scheduling : Course Research, Prof. Aijun An

Sep 2025 – Dec 2025

- Fine-tuned **Llama 3.1 8B** with **4-bit QLoRA** (Unsloth) on an NVIDIA RTX A6000, in a **graph-augmented** pipeline that converts each instance into a disjunctive graph (**PyTorch Geometric**) serialized into the prompt.
- Pre-processed the noisy **STARJOB** dataset (~130k to 9,525 usable instances), cutting fine-tuning runtime from **70h to 11.5h**.

EDUCATION & AWARDS

Master of Science in Computer Science, Specialization in Artificial Intelligence (GPA: 3.77/4)

Sep 2025 – Aug 2026

York University, Toronto, Canada

Vector Scholarship in AI, Vector Institute (\$17,500 merit award) | Supervisor: Dr. Aijun An.

Bachelor of Technology in Computer Engineering (GPA: 3.47/4)

Nov 2021 – May 2025

Pandit Deendayal Energy University, Gandhinagar, India

Publication: co-author, "Automated Sleep Stage Classification using Machine Intelligence Techniques", Springer (ADCIS 2024)